

Qatar-2b

R-0729

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_250320 · created 2026-07-28T00:50:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460754.7857 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.063
Transit depth (Rp/Rs)^2	2e-05 +/- 0.0057 [%]
Orbital Inclination (inc)	86.52 +/- 1.74
Airmass coefficient 1 (a1)	1.15 +/- 0.81
Airmass coefficient 2 (a2)	0.15 +/- 0.1
Scatter in the residuals of the lightcurve fit is	66.91 %
Best Comparison Star	#3 - [81, 185]
Optimal Aperture	2.6
Optimal Annulus	7.02
Transit Duration (day)	0.0628 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 $\pm$ 0.063 vs 0.16327 (deviation 1.83 σ)
Duration fit vs published	0.0628 d vs 0.0767 d (deviation 1.44 σ)
Mid-time O-C	3.6 min
Depth SNR	0.0
Residual scatter	66.91 %

Light curve

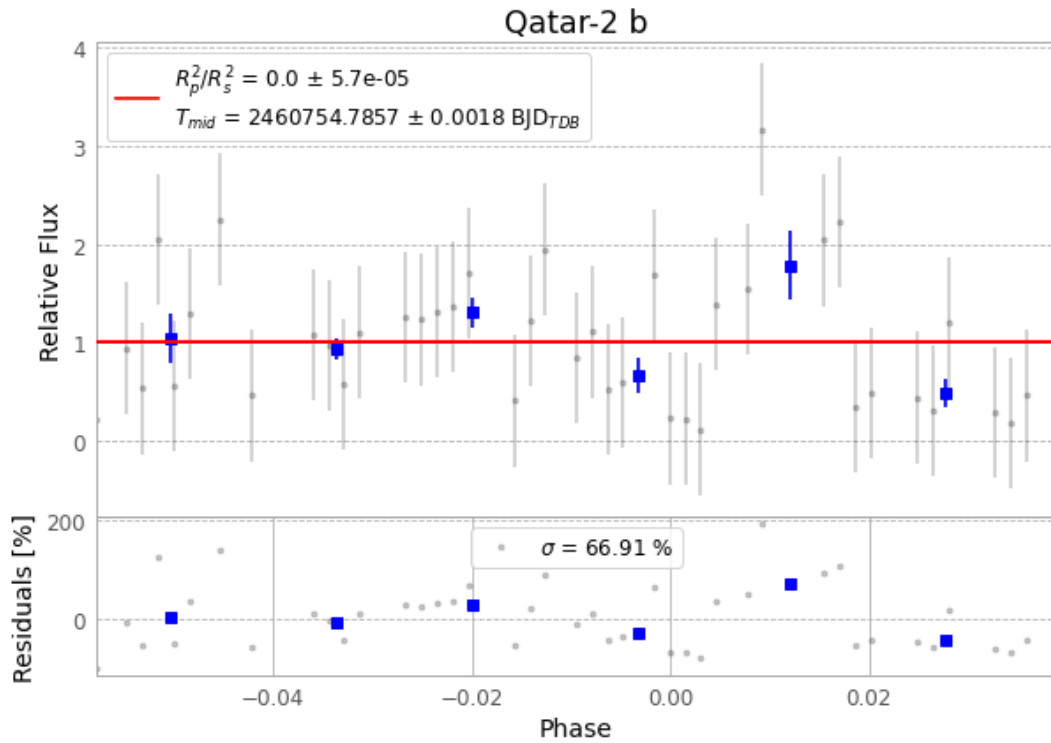


Figure 1 — FinalLightCurve_Qatar-2 b_2025-03-19.png (SHA-256 8a6968a71eaf4011...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [392, 359], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9efde2b954efc161...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

63 FITS frames (41 MB), Qatar-2250320045115.FITS → Qatar-2250320075715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-250320.log (652e1ea3c01e...), FinalLightCurve_Qatar-2 b_2025-03-19.png (8a6968a71eaf...), FinalLightCurve_Qatar-2 b_2025-03-19.csv (1cf0eac1ccf4...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 00:50Z.